WORKSHEET 6.3

From Sums to the Definite Integral

Unit 6: Integration and Accumulation of Change

Key ideas

- $\displaystyle\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\,\Delta x$, with $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$.
- To read a sum as an integral: the factor $\dfrac{b-a}{n}$ gives the width of the interval, and the expression $a + i\,\Delta x$ gives where it starts.
- A definite integral is a **signed area**: regions below the axis count as negative.
- Integrals of linear functions and semicircles can be evaluated from geometry, with no antiderivative.

Worked examples

Example 1 · A sum as an integral

Write $\displaystyle\lim_{n\to\infty}\sum_{i=1}^{n}\left(1+\frac{3i}{n}\right)^2\cdot\frac{3}{n}$ as a definite integral.

Solution. $\Delta x = \dfrac{3}{n}$, so $b-a=3$; $x_i = 1+\dfrac{3i}{n}$, so $a=1$ and $b=4$.

$\displaystyle\int_1^4 x^2\,dx$.

Example 2 · A semicircle

Evaluate $\displaystyle\int_{-2}^{2}\sqrt{4-x^2}\,dx$.

Solution. The graph is the upper half of a circle of radius 2.

$\dfrac{1}{2}\pi(2)^2 = 2\pi$.

Example 3 · An absolute value

Evaluate $\displaystyle\int_0^5 |x-3|\,dx$.

Solution. Two triangles above the axis: area $\dfrac{1}{2}(3)(3)$ on $[0,3]$ and $\dfrac{1}{2}(2)(2)$ on $[3,5]$.

$\dfrac{9}{2}+2=\dfrac{13}{2}$.

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Practice

CORE

1. Evaluate $\int_0^6 3\,dx$ using geometry.

2. Evaluate $\int_0^4 x\,dx$ using geometry.

3. Evaluate $\int_{-1}^1 \sqrt{1-x^2}\,dx$ using geometry.

4. Write a definite integral for the area under $y = x^3$ from $x = 1$ to $x = 2$.

EXAM LEVEL

5. Write $\lim_{n\to\infty} \sum_{i=1}^{n} \cos\left(\frac{3i}{n}\right) \cdot \frac{3}{n}$ as a definite integral.

6. Write $\int_1^3 \frac{1}{x}\,dx$ as the limit of a right Riemann sum.

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7. Evaluate $\int_{-2}^{4} (2 - x)\, dx$ using geometry.

8. Evaluate $\int_{0}^{3} \left(1 + \sqrt{9 - x^2}\right) dx$ using geometry.

Question 9 · Free-response practice

Let $g(x) = 2x - 4$.

(a) Find the areas of the two triangles between the graph of g and the x-axis for $0 \leq x \leq 5$.

(b) Use your answer to evaluate $\int_{0}^{5} g(x)\, dx$.

(c) Evaluate $\int_{0}^{5} |g(x)|\, dx$, and explain why it differs from part (b).

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Worked solutions

1. A rectangle: 18.
2. A triangle: $\frac{1}{2}(4)(4) = 8$.
3. A semicircle of radius 1: $\frac{\pi}{2}$.
4. $\int_1^2 x^3\,dx$.
5. $\Delta x = \frac{3}{n}$ and $x_i = \frac{3i}{n}$: $\int_0^3 \cos x\,dx$.
6. $\Delta x = \frac{2}{n}$, $x_i = 1 + \frac{2i}{n}$: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{1+\frac{2i}{n}} \cdot \frac{2}{n}$.
7. Above the axis on $[-2, 2]$: $\frac{1}{2}(4)(4) = 8$. Below on $[2, 4]$: $\frac{1}{2}(2)(2) = 2$. Value $8 - 2 = 6$.
8. A rectangle of area 3 plus a quarter circle of radius 3: $3 + \frac{9\pi}{4}$.
9. (a) Below the axis on $[0, 2]$: $\frac{1}{2}(2)(4) = 4$. Above on $[2, 5]$: $\frac{1}{2}(3)(6) = 9$.
   (b) $-4 + 9 = 5$.
   (c) $4 + 9 = 13$. The absolute value counts the region below the axis as positive.

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